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Rohan Sundeep Punamiya

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EDUCATION

Stanford University – Palo Alto, CA

- Candidate for Master of Science in Mechanical Engineering; Depth: Robotics & Kinematics ***expected June 2026***

Georgia Institute of Technology – Atlanta, GA

- Bachelor of Science in Mechanical Engineering; Minor: Computing & Theory; Concentration: Robotics ***May 2024***

EXPERIENCE

Flexiv Robotics, *R&D Robotics Engineering Intern* ***June 2025 – Sept 2025***

- Designed MPC-based whole-body controllers for simulation of future projects in Mujoco.
- Built and analyzed FE simulations for capacitive tactile sensors; designed sensor calibration pipeline and fixtures.

Proteor USA, *R&D Engineering Intern* ***June 2023 – Aug 2023***

- Designed an automated flow test and mechanical fixture to characterize over ½ million voice coil valves per year for prosthetic knee production compliance. Reduced manual testing time from 45 minutes to 20 seconds.
- Performed system ID on custom voice coil valves and assisted nonlinear controller design, enhancing steady-state response by 20% and transient response by 31%. Redesigned and tested frame/sensors with cross-functional team.

Dell Technologies, *Data Science & Decision Science Intern* ***May 2022 – Aug 2022***

- Analyzed over 2.4 million real-time data entries to determine 10 groups of failure points for product courses.
- Built automated testing program for product courses, reducing testing time from 1 week to 24 hours.

Tutoring & Academic Support, Georgia Tech, *Mentor, 1-to-1 Tutor & Physics Help Desk Tutor* ***Jan 2022 – Aug 2023***

- Advised and managed tutors, lead weekly group training meetings to improve tutor's leadership skills.
- Adapted content to 100+ students' needs, resulting in multi-grade improvements in Physics, Math, CS, and Mechanics.
- Completed CRLA International Tutor Training Program Certification (ITTPC).

RESEARCH

Exoskeleton Controller Design*, Biomechatronics Laboratory (Dr. Steve Collins)* ***Dec 2024 – Present***

- Designing a novel exoskeleton position controller for the ankle joint ; using existing torque control for stroke patients.

Bimanual Fine-Control Assembly*, IPRL (Dr. Jeannette Bohg)* ***Sept 2025 – Jan 2026***

- Training imitation and RL policies for dual arm NIST insertion/assembly and cable routing tasks.

Visual-Tactile Sensing of Object Properties*, ARMLab (Dr. Monroe Kennedy)* ***Oct 2024 – June 2025***

- Built a Gaussian Process model to fuse RGB-D images, DenseTact tactile sensing, and temperature sensing to detect fragile object mechanical properties from robot manipulator explorative procedures.

Agile Manipulation, *Lab-IDAR (Dr. Ye Zhao)* ***Dec 2021 – May 2024***

- Commanded Digit robot to demonstrate hardware box pick/place scenarios with 37% more successful human interactions, using whole-body kinematics and multi-body non-linear dynamics with real-time data.
- Designed and fabricated/prototyped a 3DOF attachable gripper with Gelsight tactile sensor embedded system.

Radiological Search Path Planning, *AREAL Laboratory (Dr. Jonathan Rogers)* ***Sept 2023 – Aug 2024***

- Optimized radiation-zone measurement path planning, increasing area covered by 44% with Voronoi tessellations.
- Built information theory-based cost function for reinforcement learning, minimizing signal interference by 26%.

LEADERSHIP

RoboJackets Club Autonomous Racing Team, *Mechanical Team Lead* ***Aug 2021 – Dec 2023***

- Supervised the design and installation of various car mechanisms, including a redesign of the braking system, car controls, and a new Ackerman steering mechanism. Each system worked smoothly in the annual AKS competition.
- Managed mechanical team and collaborated with electrical/software team, completed assembly installation on time.

SKILLS

Software:	CAD (SolidWorks, Inventor, Fusion360), COMSOL, Arduino, LabView, Gazebo, Mujoco, Genesis
Programming:	MATLAB, C/C++, Python, PyTorch, ROS2, Linux, Julia, Java, SQL, PHP, Bash, Git, BitBucket, Sourcetree
Hardware:	Waterjet, 3D Printer, Laser Cutter, Lathe, Mill, MTS Machines, ESD Testing, STM32, MSP432, myRIO
Concepts:	Deep Learning, Computer Vision, SLAM, State Estimation, Machine Design, Optimal Control